More Explanations on Grey Zones

- The "grey zones" of system context boundary in requirements engineering refer to areas where it is unclear whether something—a person, system, or aspect of the environment—is inside or outside the system's boundary.
- These zones are caused by the complexity of systems, making it difficult to define a completely precise boundary early in the process.
- Examples include not being certain if a human resources system will include a payroll module, or not knowing if an external database is relevant to the system's requirements. Navigating these zones requires careful collaboration with stakeholders to decide which elements are in scope, to minimize the grey zone's size.

<u>Why grey zones are problematic</u>

i. Ambiguity:

They create uncertainty about what is considered part of the system versus the external environment.

ii. Scope Creep:

An unclear boundary can lead to requirements being added or changed later in the development cycle, which can cause project delays and cost overruns.

iii. Inaccurate Requirements:

If aspects in the grey zone are not handled correctly, the final system may not align with its environment, leading to defects or a useless product.

Examples of grey zone elements

i. Unclear dependencies:

   It is not yet decided if a system will need to integrate with a specific third-party API or legacy system.

ii. Ambiguous user types:

   A user role might exist in the environment, but it's unclear how they will interact with the system or what their needs are.

iii. Undecided features:

   A new feature is being considered for the system, but it's not yet certain if it will be developed, leaving its functionality in the grey zone.

iv. Complex regulatory/legal environments:

   Aspects like legal compliance or industry standards may have unclear impacts, placing them in the grey zone until they are fully understood and incorporated into the requirements.

How to identify grey zones:

1. Look for Ambiguities in Boundaries
    - Check for parts of the system where it's unclear:
        - Who performs the task — the system or the user?
        - Where data originates or is processed.
        - Which system (internal or external) is responsible?
          Example: "Is report generation handled by our system or the central university portal?"

2. Review Interfaces and Interactions
    - Grey zones often appear at interfaces — where two systems, teams, or departments interact.
      Ask:
        - Are input/output responsibilities clearly defined?
        - Are there overlapping functions or duplicated roles?
          Example: Both the HR system and Payroll system store employee data — which one is the master source?
3. Identify Conflicting Assumptions
    - Different stakeholders may assume different boundaries or roles.
    - Example:
        - IT thinks maintenance is automated.
        - Operations thinks maintenance is done manually.
          → This conflicting assumption marks a grey area.